Digital Control System Analysis and Design

Third Edition

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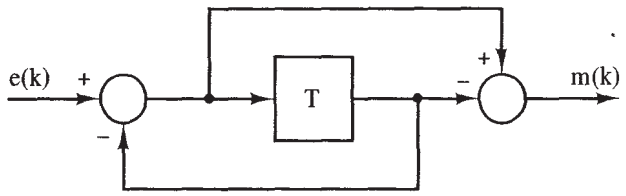


Figure 2-7 A different simulation diagram for Example 2.8.

closely related to simulation diagrams. The remainder of this chapter is devoted to the topic of state variables.

2.8 STATE VARIABLES

In preceding sections we defined a discrete-time system as one that can be described by a difference equation. If the discrete-time system is linear and time invariant, we can also represent the system by a transfer function. For a linear time-invariant discrete-time system with input $E(z)$, output $M(z)$, and transfer function $G(z)$, we can write

$$M(z) = G(z)E(z) \tag{2-44}$$

as illustrated in Section 2.7. Thus the discrete-time system may be represented by the block diagram of Figure 2-8a.

The more modern approach to the analysis and synthesis of discrete-time systems employs what is commonly called the state-variable method. In this approach the system is modeled as shown in Figure 2-8b. To be completely general, we must allow for the possibility of more than one input, and of more than one output. Thus, in Figure 2-8b, the variables $u_i(k), i = 1, \ldots, r$, are the external inputs which drive the system; the variables $y_i(k), i = 1, \ldots, p$, represent the system outputs or the system responses; and the variables $x_i(k), i = 1, \ldots, n$ are the internal or state variables of the system. The state variables represent the minimum amount of information which is necessary to determine both the future states and the system outputs for given input functions; that is, given the system states, the system dynamics and the input functions, we can determine all subsequent states and outputs.

For convenience we represent the system shown in Figure 2-8b by that shown in Figure 2-8c, where $\mathbf{u}(k)$ is the input vector, $\mathbf{y}(k)$ is the output vector, and $\mathbf{x}(k)$ is the state vector:

$$\mathbf{u}(k) = \begin{bmatrix} u_1(k) \\ u_2(k) \\ \vdots \\ u_r(k) \end{bmatrix}, \qquad \mathbf{y}(k) = \begin{bmatrix} y_1(k) \\ y_2(k) \\ \vdots \\ y_p(k) \end{bmatrix}, \qquad \mathbf{x}(k) = \begin{bmatrix} x_1(k) \\ x_2(k) \\ \vdots \\ x_n(k) \end{bmatrix}$$

The set of values that the input vector $\mathbf{u}(k)$ may assume is called the input space of the system. The output space and state space are defined in a similar manner.

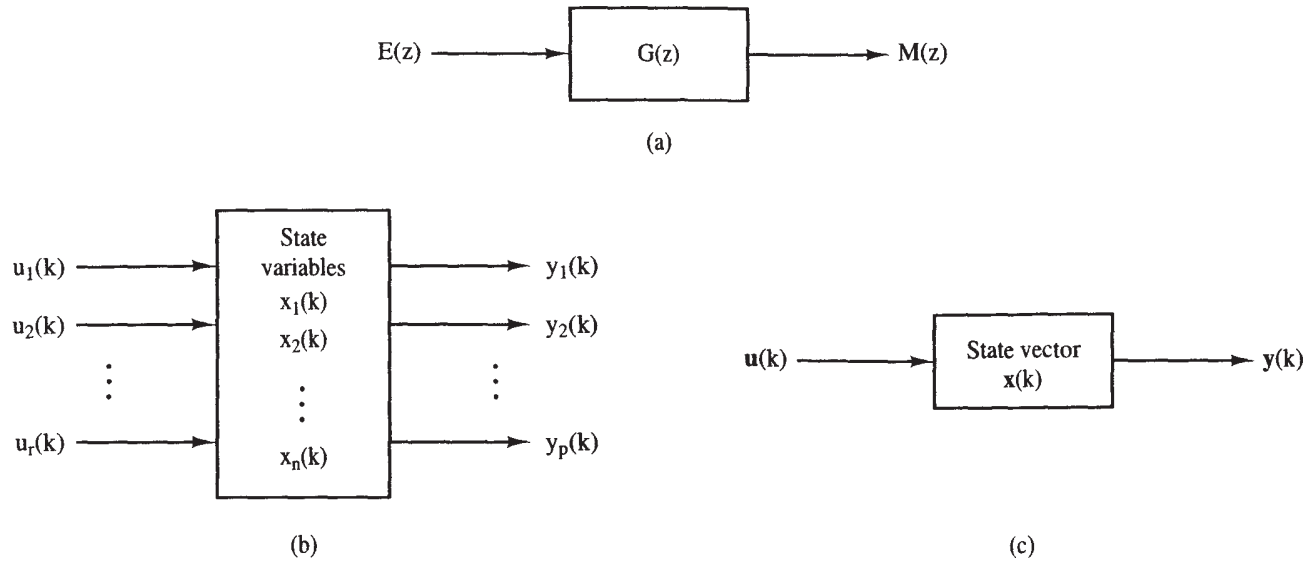


Figure 2-8 Representations of system dynamics: (a) z-transfer function representation; (b) state-variable representation; (c) state-vector representation.

In general, the equation that describes the state of the system at any time $(k + 1)$ is given by the single-valued function relationship

$$\mathbf{x}(k + 1) = \mathbf{f}[\mathbf{x}(k), \mathbf{u}(k)] \tag{2-45}$$

This equation simply states that the state $\mathbf{x}$ at time $(k + 1)$ is a function of the state and the input at the previous discrete-time increment k. The output response of the system is defined in a similar manner as

$$\mathbf{y}(k) = \mathbf{g}[\mathbf{x}(k), \mathbf{u}(k)] \tag{2-46}$$

As stated earlier, when we say time k, we actually mean time kT.

If the system is linear, then equations (2-45) and (2-46) reduce to

$$\mathbf{x}(k + 1) = \mathbf{A}(k)\mathbf{x}(k) + \mathbf{B}(k)\mathbf{u}(k) \tag{2-47}$$

$$\mathbf{y}(k) = \mathbf{C}(k)\mathbf{x}(k) + \mathbf{D}(k)\mathbf{u}(k) \tag{2-48}$$

where $\mathbf{x}(k)$ is an n-vector, $\mathbf{u}(k)$ is an r-vector, $\mathbf{y}(k)$ is a p-vector (as shown in Figure 2-8), and $\mathbf{A}(k)$, $\mathbf{B}(k)$, $\mathbf{C}(k)$, and $\mathbf{D}(k)$ are time-varying matrices of dimensions $n \times n$, $n \times r$, $p \times n$, and $p \times r$, respectively. If the system is time invariant, then the matrices in (2-47) and (2-48) are constant, and hence the equations reduce to

$$\mathbf{x}(k + 1) = \mathbf{A}\mathbf{x}(k) + \mathbf{B}\mathbf{u}(k) \tag{2-49}$$

$$\mathbf{y}(k) = \mathbf{C}\mathbf{x}(k) + \mathbf{D}\mathbf{u}(k) \tag{2-50}$$

(See Appendix IV for a review of matrices.) State-variable modeling will now be illustrated by an example.

Example 2.16

It is desired to find a state-variable model of the system described by the difference equation

$$y(k+2) = u(k) + 1.7y(k+1) - 0.72y(k)$$

Let

$$x_1(k) = y(k)$$
$$x_2(k) = x_1(k+1) = y(k+1)$$

Then

$$x_2(k+1) = y(k+2) = u(k) + 1.7x_2(k) - 0.72x_1(k)$$

or, from these equations, we write

$$x_1(k+1) = x_2(k)$$
$$x_2(k+1) = -0.72x_1(k) + 1.7x_2(k) + u(k)$$
$$y(k) = x_1(k)$$

We may express these equations in vector-matrix form of (2-49) and (2-50) as

$$\mathbf{x}(k+1) = \begin{bmatrix} 0 & 1 \\ -0.72 & 1.7 \end{bmatrix}\mathbf{x}(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix}u(k)$$
$$y(k) = [1 \quad 0]\mathbf{x}(k)$$

Equations (2-49) and (2-50) are the state equations for a linear time-invariant system and usually represent the starting point in the analysis or design of a discrete system by modern methods. However, let us first examine the connection between this approach and the z-transform method. To do this we will give one method for deriving a set of discrete state-variable equations from the z-transform transfer function.

Given the transfer function

$$G(z) = \frac{b_{n-1}z^{n-1} + b_{n-2}z^{n-2} + \cdots + b_1 z + b_0}{z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0} \tag{2-51}$$

where we have assumed that the order of the numerator is less than that of the denominator. We can write (2-51) as

$$\frac{Y(z)}{U(z)} = G(z) = \frac{b_{n-1}z^{n-1} + b_{n-2}z^{n-2} + \cdots + b_1 z + b_0}{z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0}\frac{E(z)}{E(z)} \tag{2-52}$$

where the auxiliary variable $E(z)$ has been introduced. We now let

$$Y(z) = (b_{n-1}z^{n-1} + b_{n-2}z^{n-2} + \cdots + b_1 z + b_0)E(z) \tag{2-53}$$
$$U(z) = (z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0)E(z) \tag{2-54}$$

Recall from the real translation property the correspondence

$$E(z) \rightarrow e(k)$$
$$zE(z) \rightarrow e(k+1)$$
$$z^2E(z) \rightarrow e(k+2)$$
$$\vdots$$

Under this correspondence we define the state variables

$$\begin{aligned} x_1(k) &= e(k) \\ x_2(k) &= x_1(k+1) = e(k+1) \\ x_3(k) &= x_2(k+1) = e(k+2) \\ &\vdots \\ x_n(k) &= x_{n-1}(k+1) = e(k+n-1) \end{aligned} \tag{2-55}$$

From equations (2-54) and (2-55) we obtain the state equations

$$\begin{aligned} x_1(k+1) &= x_2(k) \\ x_2(k+1) &= x_3(k) \\ x_3(k+1) &= x_4(k) \\ &\vdots \\ x_n(k+1) &= -a_0x_1(k) - a_1x_2(k) - a_2x_3(k) \cdots - a_{n-1}x_n(k) + u(k) \end{aligned} \tag{2-56}$$

which written in matrix form is

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ \vdots \\ x_n(k+1) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 & 0 & \cdots & 0 \\ & & & \vdots & & & \\ -a_0 & -a_1 & -a_2 & -a_3 & -a_4 & \cdots & -a_{n-1} \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ \vdots \\ x_n(k) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u(k) \tag{2-57}$$

or simply

$$\mathbf{x}(k+1) = \mathbf{A}\mathbf{x}(k) + \mathbf{B}u(k) \tag{2-58}$$

The output equation obtained from (2-53) is

$$y(k) = [b_0 \quad b_1 \quad \cdots \quad b_{n-1}] \begin{bmatrix} x_1(k) \\ x_2(k) \\ \vdots \\ x_n(k) \end{bmatrix} \tag{2-59}$$

or simply

$$y(k) = \mathbf{C}\mathbf{x}(k) \tag{2-60}$$

Hence equations (2-58) and (2-60) are a set of state equations for the discrete system described by equation (2-51).

Another convenient and useful representation of the discrete system is the signal flow graph or the equivalent simulation diagram. These two forms can be derived from equation (2-52)

$$G(z) = \frac{Y(z)}{U(z)} = \frac{b_{n-1}z^{-1} + b_{n-2}z^{-2} + \cdots + b_1 z^{1-n} + b_0 z^{-n}}{1 + a_{n-1}z^{-1} + \cdots + a_1 z^{1-n} + a_0 z^{-n}} \frac{E(z)}{E(z)} \tag{2-61}$$

From this expression we obtain the two equations

$$Y(z) = (b_{n-1}z^{-1} + b_{n-2}z^{-2} + \cdots + b_1 z^{1-n} + b_0 z^{-n})E(z) \tag{2-62}$$

$$U(z) = (1 + a_{n-1}z^{-1} + \cdots + a_1 z^{1-n} + a_0 z^{-n})E(z) \tag{2-63}$$

Equation (2-63) can be written as

$$E(z) = U(z) - a_{n-1}z^{-1}E(z) - \cdots - a_1 z^{1-n}E(z) - a_0 z^{-n}E(z) \tag{2-64}$$

A signal flow graph representation of the system described by equation (2-51) can be easily derived from equations (2-62) and (2-64) and is shown in Figure 2-9a.

Recall that the transfer function for a pure delay of T seconds is z^{-1}, where T is the discrete time increment (i.e., T is the interval between sampling instants $k, k+1, k+2$, etc.). Using this relationship, the signal flow graph of Figure 2-9a can be immediately converted to the equivalent simulation diagram of Figure 2-9b. We see from (2-57) and (2-59) that the states are the register outputs, as noted in Figure 2-9b.

Note the relationships that exist between equations (2-57) and (2-59) and the diagrams in Figure 2-9. One who is very familiar with the correspondence between the equations and diagrams can derive a flow graph or simulation diagram from the transfer function by inspection. From the flow graph, one may use Mason's gain formula to reconstruct the original transfer function. The structure of Figure 2-9, together with equations (2-57) and (2-59), is called either the control canonical form or the phase variable canonical form, and is useful in the design procedures presented in Chapters 9 and 10.

Another standard form that is useful in the design procedures of Chapter 9 is the observer canonical form, which is illustrated in Figure 2-10. Note, using Mason's gain formula, that the transfer function for Figure 2-10 is (2-61). It is left as an exercise for the reader to write the state equations for the observer canonical form (see Problem 2-32).

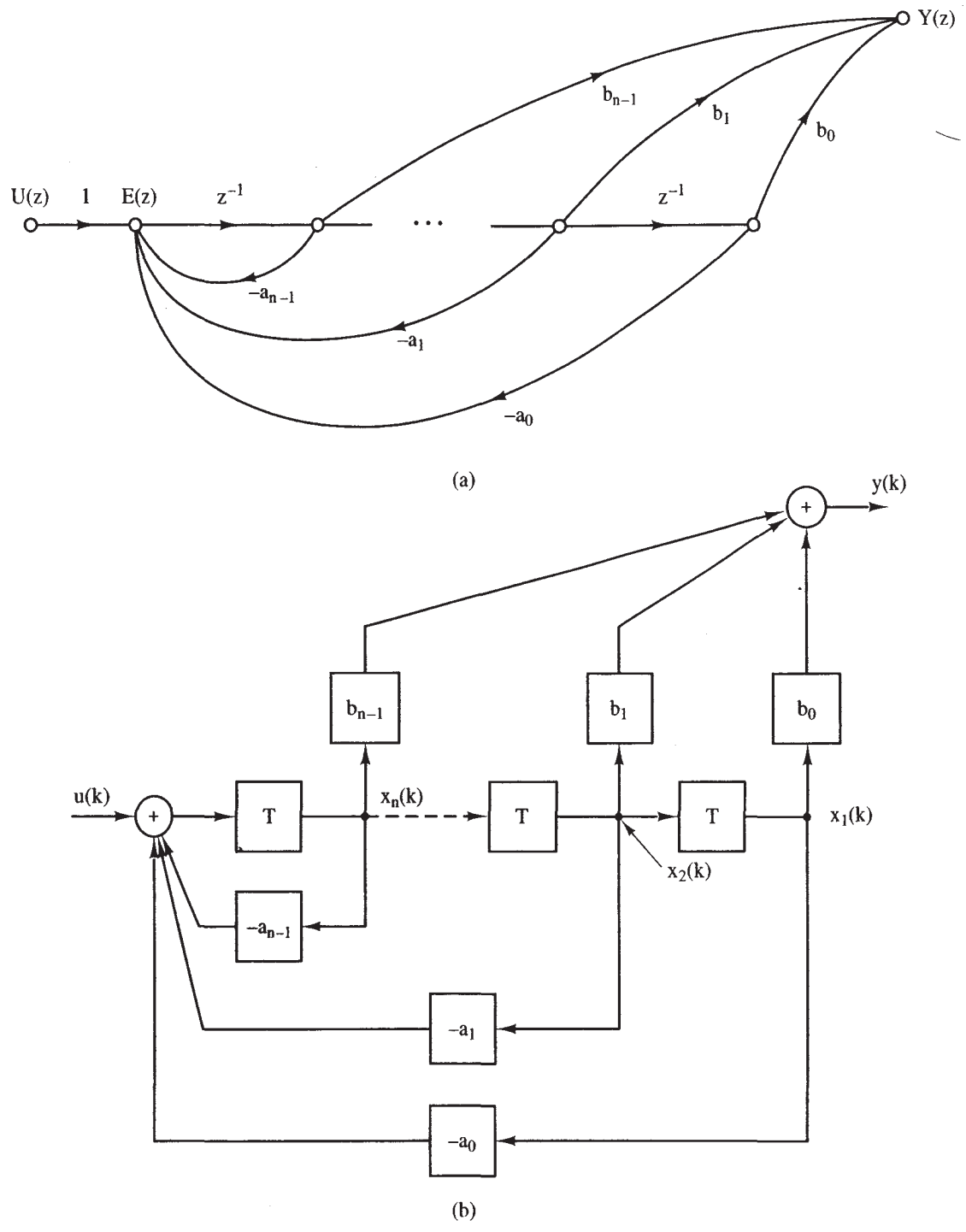


Figure 2-9 Equivalent representations of equation (2-51): (a) signal flow graph representation; (b) simulation diagram.

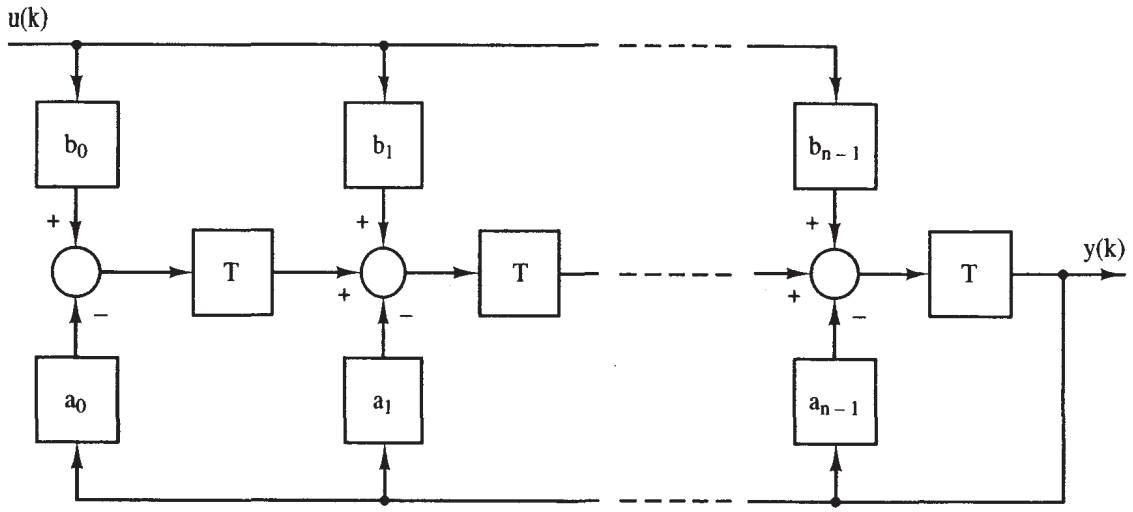


Figure 2-10 Observer canonical form.

Example 2.17

Given the following function, we want to derive a signal flow graph and corresponding state equations:

$$G(z) = \frac{Y(z)}{U(z)} = \frac{z^2 + 2z + 1}{z^3 + 2z^2 + z + \frac{1}{2}}$$

Comparing this expression with (2-51) and Figure 2-9a indicates that the signal flow graph can be derived by inspection, as shown in Figure 2-11. Note that the states are then chosen to be the delay outputs. The state equations can now be derived from Figure 2-11 or from a comparison of the given transfer function with equations (2-51), (2-57), and (2-59). The resulting state equations are

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ x_3(k+1) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -\frac{1}{2} & -1 & -2 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(k)$$

$$y(k) = \begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix}$$

The following example will demonstrate an important salient feature encountered in going from the transfer function to the state equations.

Example 2.18

Given the following transfer function, we wish to derive the state equations.

$$G(z) = \frac{Y(z)}{U(z)} = \frac{b_2 z^2 + b_1 z + b_0}{z^2 + a_1 z + a_0}$$

In this case, the order of the numerator is equal to that of the denominator. First we express the transfer function as

$$\frac{Y(z)}{U(z)} = \frac{b_2 + b_1 z^{-1} + b_0 z^{-2}}{1 + a_1 z^{-1} + a_0 z^{-2}} \frac{E(z)}{E(z)}$$

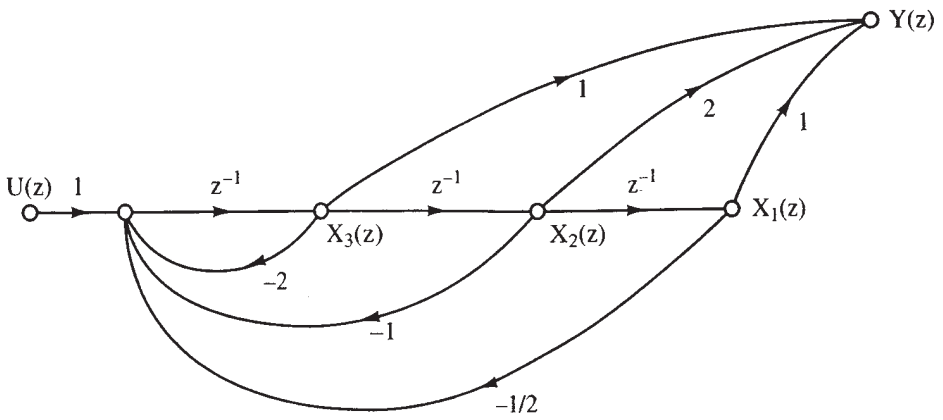


Figure 2-11 Flow graph for Example 2.17.

We see from Mason's gain formula that the signal flow graph for this transfer function is as shown in Figure 2-12. The states are shown on the flow graph, and the equations are seen to be

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -a_0 & -a_1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(k)$$

At this point it is important to note that the flow graph in Figure 2-12 is slightly different from the form presented earlier; that is, there is a direct path from $E(z)$ to $Y(z)$. From the flow graph the equation for the output is

$$Y(z) = b_0 X_1(z) + b_1 X_2(z) + b_2 E(z)$$

But

$$E(z) = U(z) - a_0 X_1(z) - a_1 X_2(z)$$

From these equations we obtain

$$Y(z) = b_2 U(z) + (b_0 - b_2 a_0) X_1(z) + (b_1 - b_2 a_1) X_2(z)$$

Hence the output equation is

$$y(k) = [(b_0 - b_2 a_0) \quad (b_1 - b_2 a_1)] \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + b_2 u(k)$$

Note the difference in this equation and (2-59).

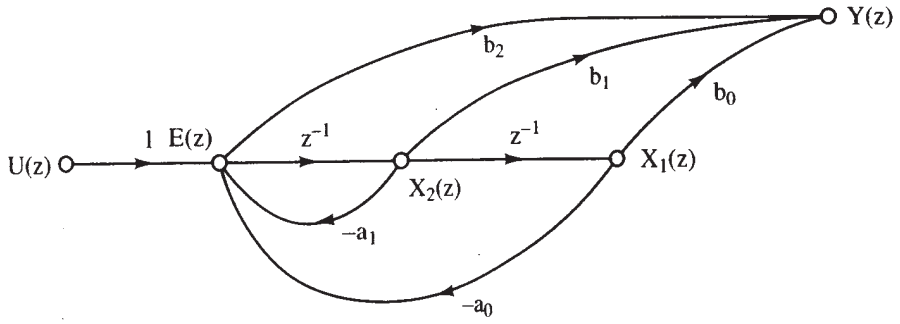


Figure 2-12 Flow graph for Example 2.18.

Whenever the numerator and denominator of the transfer function are of the same order, the output state equation must be handled as demonstrated above. Also, the output equation may be written by inspection from Figure 2-12.

Example 2.19

We wish to derive the state equations for the multivariable discrete system shown in Figure 2-13. The state equation can be obtained directly from the simulation diagram by inspection.

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \end{bmatrix} = \begin{bmatrix} 0.5 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1(k) \\ u_2(k) \end{bmatrix}$$

$$\begin{bmatrix} y_1(k) \\ y_2(k) \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1(k) \\ u_2(k) \end{bmatrix}$$

In summary, state models may be derived from transfer functions by the procedure illustrated in (2-52) through (2-59). A second procedure is:

1. Draw a simulation diagram of the system, using any convenient method.
2. Assign a state variable to each delay output.
3. Write the equation for each delay input and each system output in terms of only the delay outputs and the system input.

The system transfer function gives an input–output description of the system, while the state model gives an internal description in addition to the input–output description. The transfer function for a given system is unique, while the internal model (state model) is not.

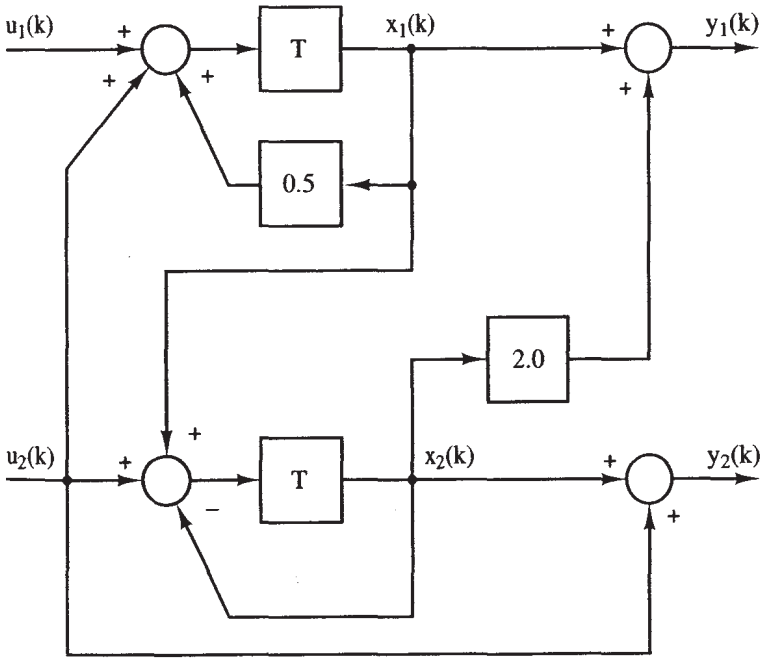


Figure 2-13 Simulation diagram for a multivariable system.

2.9 OTHER STATE-VARIABLE FORMULATIONS

We have seen in the preceding section, techniques for finding a state-variable formulation for a single-input, single-output discrete-time system, given either the system difference equation or the system transfer function. For a given system, there is no unique state-variable formulation. For a given transfer function, any simulation diagram with the system transfer function yields a valid state-variable model for the system. However, for certain analysis or design procedures, certain formulations present advantages with respect to calculations, as will be shown in later chapters. The following example illustrates some derivations of different formulations.

Example 2.20

Consider the difference equation of Example 2.16:

$$y(k+2) = u(k) + 1.7y(k+1) - 0.72y(k)$$

One state-variable model was derived in that example. Two additional models are derived here. Taking the z-transform of the difference equation yields

$$\frac{Y(z)}{U(z)} = \frac{1}{z^2 - 1.7z + 0.72}$$

This transfer function can be expressed as

$$\frac{Y(z)}{U(z)} = \left[\frac{1}{z - 0.9}\right]\left[\frac{1}{z - 0.8}\right]$$

A simulation diagram for this transfer function is given in Figure 2-14a. From this figure we write the state equations

$$\mathbf{x}(k+1) = \begin{bmatrix} 0.8 & 1 \\ 0 & 0.9 \end{bmatrix}\mathbf{x}(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix}u(k)$$

$$y(k) = [1 \quad 0]\mathbf{x}(k)$$

Also, the system transfer function can be expressed, through partial-fraction expansion, as

$$\frac{Y(z)}{U(z)} = \frac{10}{z - 0.9} + \frac{-10}{z - 0.8}$$

A simulation diagram for this transfer function is given in Figure 2-14b. From this figure we write the state equations

$$\mathbf{x}(k+1) = \begin{bmatrix} 0.9 & 0 \\ 0 & 0.8 \end{bmatrix}\mathbf{x}(k) + \begin{bmatrix} 1 \\ 1 \end{bmatrix}u(k)$$

$$y(k) = [10 \quad -10]\mathbf{x}(k)$$

In Example 2.20, we first represented a higher-order transfer function as a product of simpler transfer functions. For the general case, we would express the transfer function $G(z)$ as

$$G(z) = G_{1_c}(z)G_{2_c}(z)\cdots G_{n_c}(z) \tag{2-65}$$

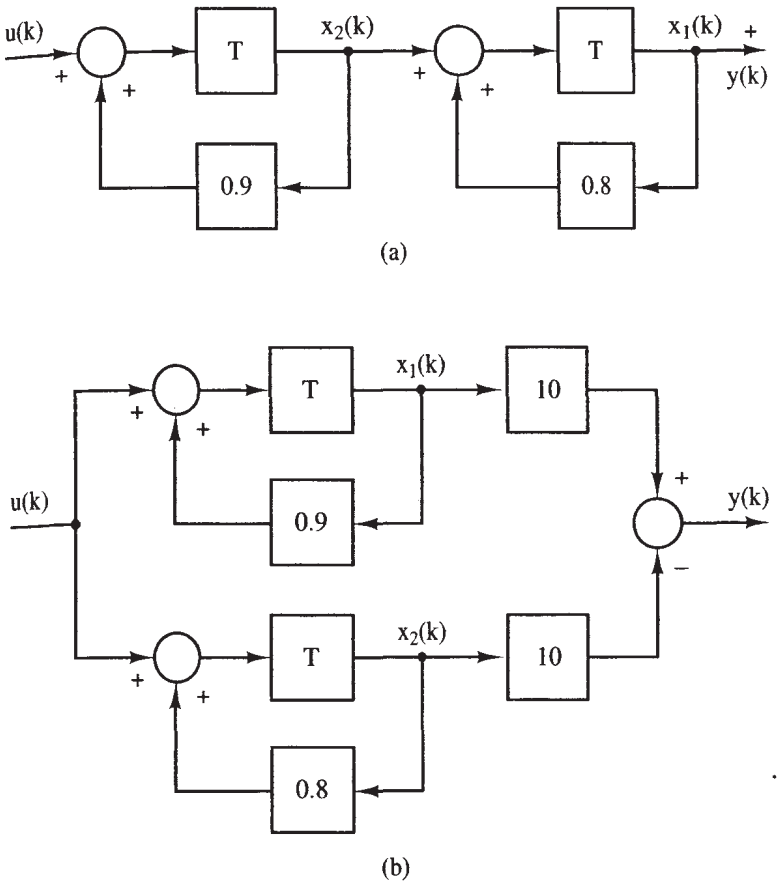


Figure 2-14 Simulation diagrams for Example 2.20.

Each of the simpler transfer functions $G_{i_c}(z)$ is realized by a technique of the preceding section, and the realizations are then connected in cascade to realize $G(z)$. If $G(z)$ contains either complex poles or complex zeros, we may choose some of the $G_{i_c}(z)$ in (2-65) to be second order, to avoid computational difficulties with complex elements in the state-variable matrices.

Next, in Example 2-20, we represented a higher-order transfer function as the sum of simpler transfer functions through partial-fraction expansion. For the general case, we would express the transfer function $G(z)$ as

$$G(z) = G_{1_p}(z) + G_{2_p}(z) + \cdots + G_{n_p}(z) \tag{2-66}$$

Each of the simpler transfer functions $G_{i_p}(z)$ is realized by a technique of Section 2.8, and the realizations are then connected in parallel to realize $G(z)$.

We have just seen three methods for arriving at different state models for a discrete-time system, with these models derived from the system transfer function. In fact, we can derive any number of different state models, given one state model of the system, through *similarity transformations*. This procedure will now be developed. Consider the state equation in the form

$$\begin{aligned} \mathbf{x}(k+1) &= \mathbf{A}\mathbf{x}(k) + \mathbf{B}\mathbf{u}(k) \\ \mathbf{y}(k) &= \mathbf{C}\mathbf{x}(k) + \mathbf{D}\mathbf{u}(k) \end{aligned} \tag{2-67}$$

Now, we can apply the linear transformation

$$\mathbf{x}(k) = \mathbf{P}\mathbf{w}(k) \tag{2-68}$$

that is,

$$x_1(k) = p_{11}w_1(k) + p_{12}w_2(k) + \cdots + p_{1n}w_n(k)$$

and so on. Hence $\mathbf{P}$ is a constant $(n \times n)$ matrix and $\mathbf{w}(k)$ is the new state vector. Note that it is necessary that $\mathbf{P}^{-1}$, the inverse of $\mathbf{P}$, exists so that $\mathbf{w}(k)$ can be determined from $\mathbf{x}(k)$. Substituting (2-68) into (2-67) yields the equations

$$\begin{aligned} \mathbf{w}(k+1) &= \mathbf{P}^{-1}\mathbf{A}\mathbf{P}\mathbf{w}(k) + \mathbf{P}^{-1}\mathbf{B}\mathbf{u}(k) \\ \mathbf{y}(k) &= \mathbf{C}\mathbf{P}\mathbf{w}(k) + \mathbf{D}\mathbf{u}(k) \end{aligned} \tag{2-69}$$

These equations can be expressed as

$$\begin{aligned} \mathbf{w}(k+1) &= \mathbf{A}_w\mathbf{w}(k) + \mathbf{B}_w\mathbf{u}(k) \\ \mathbf{y}(k) &= \mathbf{C}_w\mathbf{w}(k) + \mathbf{D}_w\mathbf{u}(k) \end{aligned} \tag{2-70}$$

where

$$\begin{aligned} \mathbf{A}_w &= \mathbf{P}^{-1}\mathbf{A}\mathbf{P}, & \mathbf{B}_w &= \mathbf{P}^{-1}\mathbf{B} \\ \mathbf{C}_w &= \mathbf{C}\mathbf{P}, & \mathbf{D}_w &= \mathbf{D} \end{aligned} \tag{2-71}$$

Thus for each different $\mathbf{P}$ for which $\mathbf{P}^{-1}$ exists, a different state model of a given system can be found.

The *characteristic equation* of a matrix $\mathbf{A}$ is defined by the determinant [8]

$$|z\mathbf{I} - \mathbf{A}| = 0 \tag{2-72}$$

and the *characteristic values* (or *eigenvalues*) of the matrix are the roots of the characteristic equation; that is, z_i are the characteristic values of $\mathbf{A}$ if

$$|z\mathbf{I} - \mathbf{A}| = (z - z_1)(z - z_2)\cdots(z - z_n) = 0 \tag{2-73}$$

It will be shown in Chapter 7 that (2-73) determines system stability for the system of (2-67). Note that the characteristic equation of the system matrix is unchanged through the linear transformation,

$$\begin{aligned} |z\mathbf{I} - \mathbf{A}_w| &= |z\mathbf{I} - \mathbf{P}^{-1}\mathbf{A}\mathbf{P}| = |z\mathbf{P}^{-1}\mathbf{I}\mathbf{P} - \mathbf{P}^{-1}\mathbf{A}\mathbf{P}| \\ &= |\mathbf{P}^{-1}||z\mathbf{I} - \mathbf{A}||\mathbf{P}| \\ &= |z\mathbf{I} - \mathbf{A}| \end{aligned} \tag{2-74}$$

A linear transformation of the form

$$\mathbf{A}_w = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} \tag{2-75}$$

which is called a *similarity transformation*, has the following properties:

1. As shown above, the characteristic values of the matrix are unchanged under the transformation.
2. The determinant of $\mathbf{A}_w$ (which is the product of characteristic values) is equal to the determinant of $\mathbf{A}$:

$$|\mathbf{A}_w| = |\mathbf{P}^{-1}\mathbf{A}\mathbf{P}| = |\mathbf{P}^{-1}||\mathbf{A}||\mathbf{P}| = |\mathbf{A}| = z_1 z_2 \cdots z_n$$

3. The trace of $\mathbf{A}_w$ is equal to the trace of $\mathbf{A}$:

$$\text{tr}\,\mathbf{A}_w = \text{tr}\,\mathbf{A} = z_1 + z_2 + \cdots + z_n$$

since the trace of a matrix, which is the sum of the diagonal elements, is equal to the sum of its characteristic values. The latter property follows immediately from the first property.

4. As is shown in Section 2.10,

$$\mathbf{C}[z\mathbf{I} - \mathbf{A}]^{-1}\mathbf{B} + \mathbf{D} = \mathbf{C}_w[z\mathbf{I} - \mathbf{A}_w]^{-1}\mathbf{B}_w + \mathbf{D}_w$$

Example 2.21

For the system of Example 2.20, one state-variable model given is

$$\mathbf{x}(k+1) = \begin{bmatrix} 0.8 & 1 \\ 0 & 0.9 \end{bmatrix}\mathbf{x}(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix}u(k)$$

$$y(k) = [1 \quad 0]\mathbf{x}(k)$$

We arbitrarily choose a linear-transformation matrix

$$\mathbf{P} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

The inverse of P is given by

$$\mathbf{P}^{-1} = \frac{[\text{Cof}[\mathbf{P}]]^T}{|\mathbf{P}|}$$

where $[\,\cdot\,]^T$ indicates the transpose, and Cof[P] denotes the matrix of cofactors of P. Thus

$$\text{Cof}[\mathbf{P}] = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

and

$$|\mathbf{P}| = 2$$

Then

$$\mathbf{P}^{-1} = \begin{bmatrix} 0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix}$$

Thus, from (2-71),

$$\mathbf{A}_w = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{bmatrix} 0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix}\begin{bmatrix} 0.8 & 1 \\ 0 & 0.9 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix}\begin{bmatrix} 1.8 & 0.2 \\ 0.9 & 0.9 \end{bmatrix} = \begin{bmatrix} 1.35 & 0.55 \\ -0.45 & 0.35 \end{bmatrix}$$

$$\mathbf{B}_w = \mathbf{P}^{-1}\mathbf{B} = \begin{bmatrix} 0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$$

$$\mathbf{C}_w = \mathbf{C}\mathbf{P} = [1 \quad 0]\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = [1 \quad -1]$$

Then the new state equations are

$$\mathbf{w}(k+1) = \begin{bmatrix} 1.35 & 0.55 \\ -0.45 & 0.35 \end{bmatrix}\mathbf{w}(k) + \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}u(k)$$

$$y(k) = [1 \quad -1]\mathbf{w}(k)$$

Note that

$$|z\mathbf{I} - \mathbf{A}| = \begin{vmatrix} z - 0.8 & -1 \\ 0 & z - 0.9 \end{vmatrix} = z^2 - 1.7z + 0.72 = (z - 0.8)(z - 0.9)$$

and

$$|z\mathbf{I} - \mathbf{A}_w| = \begin{vmatrix} z - 1.35 & -0.55 \\ 0.45 & z - 0.35 \end{vmatrix} = z^2 - 1.7z + 0.72$$

Hence the characteristic values are $z_1 = 0.8$ and $z_2 = 0.9$. Also,

$$|\mathbf{A}| = |\mathbf{A}_w| = 0.72 = z_1 z_2$$

and

$$\operatorname{tr}\mathbf{A} = \operatorname{tr}\mathbf{A}_w = 1.7 = z_1 + z_2$$

The results of this example can be verified with the MATLAB program CTRL, described in Appendix VII.

If a system has distinct characteristic values, we may derive a state-variable model in which the system matrix is diagonal. Consider a vector $\mathbf{m}_i$ and a scalar z_i defined by the equation

$$\mathbf{A}\mathbf{m}_i = z_i\mathbf{m}_i \tag{2-76}$$

where $\mathbf{m}_i = [m_{1i} \quad m_{2i} \quad \cdots \quad m_{ni}]^T$. We express (2-76) as

$$(z_i\mathbf{I} - \mathbf{A})\mathbf{m}_i = \mathbf{0}$$

For a nontrivial solution for this equation to exist, it is required that

$$|z_i\mathbf{I} - \mathbf{A}| = 0$$

Hence, from (2-72), it is seen that z_i is a characteristic value (also called an eigen-

value) of **A**. The vector $\mathbf{m}_i$ is called a *characteristic vector*, or *eigenvector*, of **A**. From (2-76) we can construct the equation

$$\mathbf{A}[\mathbf{m}_1 \quad \mathbf{m}_2 \quad \cdots \quad \mathbf{m}_n] = [\mathbf{m}_1 \quad \mathbf{m}_2 \quad \cdots \quad \mathbf{m}_n]\begin{bmatrix} z_1 & 0 & \cdot & 0 \\ 0 & z_2 & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & \cdot & z_n \end{bmatrix}$$

or

$$\mathbf{AM} = \mathbf{M}\Lambda$$

where **M**, called the *modal matrix*, is composed of the characteristic vectors as columns, and Λ is a diagonal matrix with the characteristic values of **A** as the diagonal elements. The characteristic vectors are linearly independent provided the characteristic values are distinct. Hence we can write

$$\Lambda = \mathbf{M}^{-1}\mathbf{AM} \tag{2-77}$$

Then, in the general similarity transformation, **P** is equal to **M**, the modal matrix. An example will be given to illustrate this procedure.

Example 2.22



For the system of Example 2.21

$$\mathbf{x}(k+1) = \begin{bmatrix} 0.8 & 1 \\ 0 & 0.9 \end{bmatrix}\mathbf{x}(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix}u(k)$$

$$y(k) = [1 \quad 0]\mathbf{x}(k)$$

From Example 2.21, the characteristic values are $z_1 = 0.8$ and $z_2 = 0.9$. From (2-76),

$$\begin{bmatrix} 0.8 & 1 \\ 0 & 0.9 \end{bmatrix}\begin{bmatrix} m_{11} \\ m_{21} \end{bmatrix} = 0.8\begin{bmatrix} m_{11} \\ m_{21} \end{bmatrix}$$

Thus

$$0.8m_{11} + m_{21} = 0.8m_{11}$$

$$0.9m_{21} = 0.8m_{21}$$

Hence $m_{21} = 0$ and m_{11} is arbitrary. Let $m_{11} = 1$. Also,

$$\begin{bmatrix} 0.8 & 1 \\ 0 & 0.9 \end{bmatrix}\begin{bmatrix} m_{12} \\ m_{22} \end{bmatrix} = 0.9\begin{bmatrix} m_{12} \\ m_{22} \end{bmatrix}$$

Then m_{22} is seen to be arbitrary; let $m_{22} = 1$. Then $m_{12} = 10$. The modal matrix and its inverse are

$$\mathbf{M} = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} = \begin{bmatrix} 1 & 10 \\ 0 & 1 \end{bmatrix}$$

and

$$\mathbf{M}^{-1} = \begin{bmatrix} 1 & -10 \\ 0 & 1 \end{bmatrix}$$

From (2-71) and (2-77),

$$\mathbf{\Lambda} = \mathbf{M}^{-1}\mathbf{A}\mathbf{M} = \begin{bmatrix} 0.8 & 0 \\ 0 & 0.9 \end{bmatrix}$$

$$\mathbf{B}_w = \mathbf{M}^{-1}\mathbf{B} = \begin{bmatrix} -10 \\ 1 \end{bmatrix}$$

$$\mathbf{C}_w = \mathbf{C}\mathbf{M} = [1 \quad 10]$$

Hence the new state model is given by

$$\mathbf{w}(k+1) = \begin{bmatrix} 0.8 & 0 \\ 0 & 0.9 \end{bmatrix}\mathbf{w}(k) + \begin{bmatrix} -10 \\ 1 \end{bmatrix}u(k)$$

$$y(k) = [1 \quad 10]\mathbf{w}(k)$$

A MATLAB program that calculates the characteristic vectors and the characteristic values is given by

```
A = [0.8  1;  0  0.9];
[V,D] = eig(A)
```

result: V: 1 0.995 D: 0.8 0
0 0.0995 0 0.9

The matrix V gives the characteristic vectors as columns (each determined only within a constant factor), and D gives the characteristic values on its diagonal.

2.10 TRANSFER FUNCTIONS

In the techniques described earlier in this text for obtaining a state-variable formulation, we derived the state model from the transfer function. In this section we demonstrate how one may obtain the transfer function, given the state model.

As a first approach, given the state equations of a discrete-time system, we can construct a simulation diagram. The transfer function can then be obtained from the simulation diagram, using Mason's gain formula.

Example 2.23

Consider the state model derived in Example 2.21.

$$\mathbf{x}(k+1) = \begin{bmatrix} 1.35 & 0.55 \\ -0.45 & 0.35 \end{bmatrix}\mathbf{x}(k) + \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}u(k)$$

$$y(k) = [1 \quad -1]\mathbf{x}(k)$$

A simulation diagram, constructed from these equations, is shown in Figure 2-15.

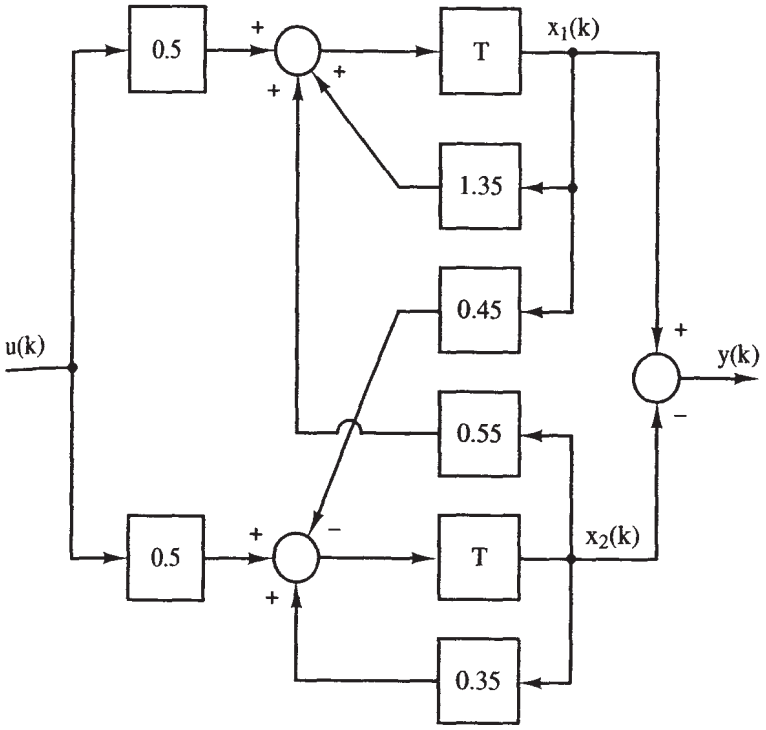


Figure 2-15 System for Example 2.23.

Application of Mason's gain formula to this figure yields

$$\frac{Y(z)}{U(z)} = \frac{0.5z^{-1}(1 - 0.35z^{-1}) - 0.5z^{-1}(1 - 1.35z^{-1}) + (0.5)(0.45)z^{-2} + (0.5)(0.55)z^{-2}}{1 - 1.35z^{-1} - 0.35z^{-1} + (0.45)(0.55)z^{-2} + (1.35z^{-1})(0.35z^{-1})}$$

$$= \frac{z^{-2}}{1 - 1.7z^{-1} + 0.72z^{-2}} = \frac{1}{z^2 - 1.7z + 0.72}$$

Since this is the system transfer function, we see that the derived state model is valid.

Another approach for obtaining the system transfer function from the state equations is to take the z-transform of the state equations and eliminate $\mathbf{X}(z)$. Since

$$\mathbf{x}(k + 1) = \mathbf{A}\mathbf{x}(k) + \mathbf{B}u(k) \tag{2-78}$$

taking the z-transform yields

$$z\mathbf{X}(z) - z\mathbf{x}(0) = \mathbf{A}\mathbf{X}(z) + \mathbf{B}U(z) \tag{2-79}$$

Since, in deriving transfer functions, we ignore initial conditions, (2-79) can be expressed as

$$[z\mathbf{I} - \mathbf{A}]\mathbf{X}(z) = \mathbf{B}U(z) \tag{2-80}$$

Solving for $\mathbf{X}(z)$, we obtain

$$\mathbf{X}(z) = [z\mathbf{I} - \mathbf{A}]^{-1}\mathbf{B}U(z) \tag{2-81}$$

Also, since

$$y(k) = \mathbf{C}\mathbf{x}(k) + Du(k) \tag{2-82}$$

then

$$Y(z) = \mathbf{C}\mathbf{X}(z) + DU(z) \tag{2-83}$$

Substituting (2-81) into (2-83) yields

$$Y(z) = [\mathbf{C}[z\mathbf{I} - \mathbf{A}]^{-1}\mathbf{B} + D]U(z)$$

The system transfer function is then seen to be

$$G(z) = \mathbf{C}[z\mathbf{I} - \mathbf{A}]^{-1}\mathbf{B} + D \tag{2-84}$$

This technique will now be illustrated by an example.

Example 2.24



Consider again the state equations of Example 2.23.

$$\mathbf{x}(k+1) = \begin{bmatrix} 1.35 & 0.55 \\ -0.45 & 0.35 \end{bmatrix}\mathbf{x}(k) + \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}u(k)$$

$$y(k) = [1 \quad -1]\mathbf{x}(k)$$

Now

$$[z\mathbf{I} - \mathbf{A}] = \begin{bmatrix} z - 1.35 & -0.55 \\ 0.45 & z - 0.35 \end{bmatrix}$$

Thus

$$|z\mathbf{I} - \mathbf{A}| = z^2 - 1.7z + 0.72$$

Also,

$$\text{Cof}[z\mathbf{I} - \mathbf{A}] = \begin{bmatrix} z - 0.35 & -0.45 \\ 0.55 & z - 1.35 \end{bmatrix}$$

Then

$$[z\mathbf{I} - \mathbf{A}]^{-1} = \frac{[\text{Cof}[z\mathbf{I} - \mathbf{A}]]^T}{|z\mathbf{I} - \mathbf{A}|} = \frac{1}{z^2 - 1.7z + 0.72}\begin{bmatrix} z - 0.35 & 0.55 \\ -0.45 & z - 1.35 \end{bmatrix}$$

From (2-84), since $D = 0$,

$$\begin{aligned} G(z) &= \mathbf{C}[z\mathbf{I} - \mathbf{A}]^{-1}\mathbf{B} \\ &= \frac{1}{z^2 - 1.7z + 0.72}[1 \quad -1]\begin{bmatrix} z - 0.35 & 0.55 \\ -0.45 & z - 1.35 \end{bmatrix}\begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} \\ &= \frac{1}{z^2 - 1.7z + 0.72}[1 \quad -1]\begin{bmatrix} 0.5z + 0.1 \\ 0.5z - 0.9 \end{bmatrix} \\ &= \frac{1}{z^2 - 1.7z + 0.72} \end{aligned}$$

This, of course, has been shown to be the transfer function.

Another property of the similarity transformation, applied to state models as given in (2-70), is seen from the example above. Since the transfer function of a system is invariant under a similarity transformation, then, from (2-70) and (2-84),

$$\mathbf{C}[z\mathbf{I} - \mathbf{A}]^{-1}\mathbf{B} + \mathbf{D} = \mathbf{C}_w[z\mathbf{I} - \mathbf{A}_w]^{-1}\mathbf{B}_w + \mathbf{D}_w \tag{2-85}$$

The proof of this property is left as an exercise for the reader (see Problem 2-35).

In this section two techniques are presented for obtaining the transfer function from the state equations. For high-order systems, the second method is more attractive. The second technique, as given in (2-84), can be implemented as a computer program [9]. For example, a MATLAB implementation of (2-84) for Example 2.24 is

```
A = [1.35  0.55;  −0.45  0.35];
B = [0.5;  0.5];
C = [1  −1];
D = 0;
[n,d] = ss2tf(A,  B,  C,  D)

result:  n = 0.0    0.0   1.0
         d = 1.0   −1.7   0.72
```

2.11 SOLUTIONS OF THE STATE EQUATIONS

In this section we develop the general solution of linear time-invariant state equations. It will be seen that the key to the solution of the state equations is the calculation of the state transition matrix. Two related techniques, based on the z-transform, for calculating the state transition matrix are presented. Then the solution of state equations via the digital computer is mentioned.

Recursive Solution

We will first assume that the system is time invariant (fixed) and that $\mathbf{x}(0)$ and $\mathbf{u}(j), j = 0, 1, 2, \ldots$, are known. Now the system equations are

$$\mathbf{x}(k+1) = \mathbf{A}\mathbf{x}(k) + \mathbf{B}\mathbf{u}(k) \tag{2-86}$$

$$\mathbf{y}(k) = \mathbf{C}\mathbf{x}(k) + \mathbf{D}\mathbf{u}(k) \tag{2-87}$$

In a recursive manner it is obvious that

$$\mathbf{x}(1) = \mathbf{A}\mathbf{x}(0) + \mathbf{B}\mathbf{u}(0)$$

and

$$\mathbf{x}(2) = \mathbf{A}\mathbf{x}(1) + \mathbf{B}\mathbf{u}(1)$$